$$M_X(t) = (1-2t)^{-\frac{n}{2}}, \quad t < \frac{1}{2}$$

$$\frac{\partial M_X(t)}{\partial t} = n(1-2t)^{-\frac{n}{2}-1}$$

$$\left.\frac{d M_X(t)}{dt}\right|_{t=0} = n$$

$$\Rightarrow E(X) = n$$

$$\frac{d^2 M_X(t)}{dt^2} = \frac{n(n+2)}{(1-2t)^{\frac{n}{2}}(2t-1)^2}$$

$$E(X^2) = \left.\frac{d^2 M_X(t)}{dt^2}\right|_{t=0} = n(n+2$$

$$\begin{aligned} Var(X) &= n(n+2) - n^2 \\ &= n^2 + 2n - n^2 \\ &= 2n \end{aligned}$$

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